ChatGPT said:

Date: 6 November 2026

Product Requirements Document: Fitness Assessment, Personalization & Recommendation Platform

Product definition

Unified Fitness Platform

Product foundation: MyFitnessPal + Strava + LADDER

Target market: Everyone, from complete beginners to experienced athletes.

Recommendation model: Hybrid, combining intelligent recommendations with user-controlled decisions.

Core product principle: Assess → Understand → Recommend → Act → Measure → Adapt.

1. Product vision

The platform will provide a personalized fitness experience that helps individuals understand their current fitness position, establish achievable goals, discover appropriate physical activities, follow structured training programs, monitor nutrition and track progress.

Rather than requiring users to understand fitness programming themselves, the platform will guide them through a structured assessment and generate a personalized fitness blueprint.

The platform combines three connected capabilities:

Product inspiration	Capability adopted
MyFitnessPal	Nutrition, calorie tracking, food diary and meal planning
Strava	Outdoor activities, GPS tracking, social fitness and challenges
LADDER	Structured training programs, exercise instructions and progressive training

The differentiation is the connection between these capabilities.

For example, a person who wants to lose weight should not receive only a calorie target. They should receive a coordinated nutrition strategy, appropriate physical activities, a weekly workout schedule and measurable progress indicators.

1.1 Product objectives

1. Help users identify realistic and measurable fitness goals.
2. Classify users according to fitness experience, needs and preferences.
3. Recommend appropriate physical activities and training programs.
4. Connect exercise, nutrition and progress tracking.
5. Encourage consistency through accountability and community.
6. Adapt recommendations as users improve or their circumstances change.
7. Create a sustainable subscription-based fitness business.

1.2 Target user segments

The platform must support different user segments without creating separate products.

Segment	Main need	Product response
Complete beginners	Know where to start	Beginner assessment and guided plans
Weight management users	Manage weight sustainably	Nutrition and activity plans
Strength enthusiasts	Build muscle and strength	Progressive resistance programs
Runners and cyclists	Improve performance	GPS tracking and endurance programs
General wellness users	Stay healthy and active	Walking, mobility and consistency plans
Home fitness users	Exercise without gym access	Equipment-free and home programs
Experienced athletes	Optimize performance	Advanced analytics and structured training
Older adults	Maintain functional capacity	Appropriate mobility, balance and strength activities

Users with limitations	Find suitable activities	Health-aware recommendations and professional referrals where necessary

Important scope boundary: Everyone is the intended audience, not permission to prescribe the same training approach to everyone. Minors, older adults, pregnant or postpartum users, and users with health conditions need appropriate safeguards and specialized recommendation rules.

2. Fitness assessment system

Core product capability

The assessment system is the entry point into the personalization engine.

It should collect enough information to make meaningful recommendations without overwhelming users.

2.1 Assessment design principles

- Complete the standard assessment in approximately 3–5 minutes.
- Use conversational questions and visual answer cards.
- Support single-selection and multiple-selection questions.
- Allow users to skip nonessential questions.
- Explain why sensitive or optional information is requested.
- Save responses as the user progresses.
- Allow users to edit their answers later.
- Separate health screening from fitness preference questions.
- Generate an initial fitness blueprint immediately after completion.
- Never confuse a user's aspirations with their current physical capabilities.

2.2 Detailed assessment questionnaire

I recommend six assessment stages.

Stage A: Personal profile

Objective: Establish basic demographic and physical context.

ID	Question	Input	Required
A01	What is your date of birth?	Date picker	Yes
A02	What is your biological sex?	Female, Male, Prefer not to say	Optional
A03	What is your height?	cm or ft/in	Optional
A04	What is your current weight?	kg or lbs	Optional
A05	What country do you live in?	Country selector	Yes
A06	What measurement system do you prefer?	Metric / Imperial	Yes

Design considerations:

- Calculate age from date of birth.
- Use age to determine the appropriate assessment and safety pathway.
- Height and weight should be optional unless needed for a selected goal.
- Never infer body composition from BMI alone.
- Users should be able to use the app without disclosing sensitive demographic information unnecessarily.

Stage B: Fitness goals

Objective: Understand what the user wants to achieve.

B01. What is your primary fitness goal?

Single selection.

Goal	Description
Weight management	Manage weight and body composition
Build muscle	Increase muscle mass

Build strength	Improve physical strength
Improve endurance	Improve cardiovascular capacity
Improve flexibility	Improve mobility and movement
Improve general fitness	Improve overall physical fitness
Maintain a healthy lifestyle	Establish sustainable habits
Sports performance	Improve performance in a particular sport
Prepare for an event	Prepare for a race, competition or other event

B02. Do you have any secondary fitness goals?

Multiple selection.

- Improve cardiovascular health.
- Build muscle.
- Improve strength.
- Improve endurance.
- Improve flexibility.
- Improve balance and coordination.
- Improve sleep and recovery habits.
- Increase daily physical activity.
- Improve consistency.
- Improve athletic performance.

B03. What would success look like to you?

Free-text response with suggested examples.

- Lose a specified amount of weight.
- Run 5 km continuously.
- Complete 20 push-ups.
- Train four times weekly.
- Improve strength in major lifts.
- Complete a cycling event.

B04. Do you have a target date?

- Within 4 weeks.
- Within 8 weeks.
- Within 12 weeks.
- Within 6 months.
- No specific deadline.

The system must distinguish a user-selected target from a medically or physiologically validated expectation.

Stage C: Current fitness condition

Objective: Understand the user's starting point.

ID	Question	Options
C01	How would you describe your current fitness level?	Beginner, Intermediate, Advanced, Not sure
C02	How frequently do you exercise?	Never, 1–2 times/week, 3–4 times/week, 5+ times/week
C03	How long have you exercised consistently?	Just starting, Under 3 months, 3–6 months, 6–12 months, 1–3 years, 3+ years
C04	What activities do you currently engage in?	Walking, running, cycling, gym, home workouts, swimming, yoga, sports, other
C05	How active is your typical day?	Mostly sitting, Some movement, Frequently standing/walking, Physically demanding
C06	How confident are you about exercising independently?	Not confident, Somewhat confident, Confident, Very confident

C01 is self-reported. It must not be treated as an objective fitness assessment.

A user who claims to be advanced but reports no recent exercise should receive a conservative starting recommendation.

Stage D: Lifestyle and training preferences

Objective: Determine what activities are practical for the user.

ID	Question	Options
D01	How many days can you exercise weekly?	1–2, 3, 4–5, 6–7
D02	How much time can you dedicate per session?	10–15, 15–30, 30–45, 45–60, 60+ minutes
D03	Where do you prefer exercising?	Home, Gym, Outdoors, Combination
D04	What equipment can you access?	None, Dumbbells, Resistance bands, Home equipment, Full gym
D05	Which activities do you enjoy?	Walking, running, cycling, strength training, swimming, yoga, Pilates, sports, group workouts
D06	Which activities would you rather avoid?	Same activity options, plus none
D07	Do you prefer exercising alone or with others?	Alone, With friends, Group classes, No preference
D08	When do you prefer exercising?	Morning, Afternoon, Evening, Flexible

The engine should treat these preferences as recommendation constraints, not merely profile information.

Stage E: Health and safety screening

Safety-critical assessment

Objective: Identify situations where standard exercise recommendations may be unsuitable.

ID	Question	Options
E01	Do you have an injury or physical limitation that affects movement?	Yes, No, Not sure
E02	Do you have a medical condition that could affect exercise?	Yes, No, Prefer not to say
E03	Have you been advised by a healthcare professional to restrict physical activity?	Yes, No, Not sure
E04	Are you currently recovering from an illness, injury or surgery?	Yes, No
E05	Do you experience symptoms during exercise that concern you, such as chest pain, fainting or unusual breathlessness?	Yes, No, Prefer not to say
E06	Is there anything else you want us to consider when recommending activities?	Optional free text

Conditional follow-up questions:

- If E01 is yes, ask which movements or activities are affected.
- If E02 is yes, optionally collect a condition category and relevant exercise restrictions.
- If E03 is yes, ask whether professional guidance has been obtained.
- If E04 is yes, ask whether the user has been cleared to resume exercise.
- If E05 is yes, pause exercise recommendations and display appropriate medical guidance.

Do not attempt to diagnose conditions or use an automated questionnaire as medical clearance.

For urgent warning symptoms, the application must advise the user to stop exercising and seek appropriate medical attention.

For users with unresolved health restrictions, the system should limit recommendations to appropriate general information until professional guidance is obtained.

Stage F: Motivation and commitment

Objective: Understand what will help the user remain consistent.

ID	Question	Options

F01	What motivates you to exercise?	Health, Appearance, Performance, Energy, Confidence, Competition, Social interaction
F02	What usually prevents you from exercising?	Time, Motivation, Cost, Lack of knowledge, Equipment, Access to facilities, Physical limitations
F03	How much guidance do you prefer?	Minimal guidance, Some guidance, Step-by-step coaching
F04	Would you like reminders?	Yes, No, Custom schedule
F05	How would you describe your commitment?	Getting started, Building consistency, Moderately committed, Highly committed

This section is particularly useful for retention and engagement personalization.

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3. Assessment scoring and classification logic

The classification engine should produce a structured fitness profile rather than assign a single permanent category.

3.1 Classification dimensions

Each user receives six independent classifications.

Dimension	Possible classifications
Primary goal	Weight management, muscle growth, endurance, mobility, general fitness, sports performance
Experience	Beginner, intermediate, advanced
Activity baseline	Sedentary, low active, moderately active, highly active
Training environment	Home, gym, outdoor, mixed
Availability	Low, moderate, high
Safety status	Standard, needs modification, professional guidance required, exercise paused pending assessment

The system should also maintain preferences, equipment access and secondary goals.

3.2 Fitness pathways

The initial recommendation engine will support six pathways.

Pathway	Main training focus	Nutrition focus
Weight management	Walking, cardio, resistance training	Sustainable energy balance
Muscle development	Progressive resistance training	Adequate energy and protein
Strength development	Structured strength programs	Recovery and adequate nutrition
Endurance development	Running, cycling and aerobic training	Training-supportive nutrition
Mobility and functional fitness	Mobility, balance and appropriate strength	General balanced nutrition
Healthy lifestyle	Walking, accessible exercise and habit formation	Consistent balanced eating

Sports performance can operate as a specialized pathway using sport-specific training templates.

A user may belong to more than one pathway, with one primary goal and several supporting goals.

3.3 Scoring model

I recommend a weighted scoring model for activity recommendations.

Each candidate activity is evaluated against the user's goals and constraints.

Scoring dimension	Weight
Alignment with primary goal	30%
User preference	20%
Fitness experience suitability	15%

Available equipment	10%
Available time	10%
Activity environment	10%
Secondary goal alignment	5%
Total	100%

The score is:

$$S(a,u) = \sum_{i=1}^{7} w_i \cdot r_i(a,u)$$

Where:

- $S(a,u)$ = suitability score of activity a for user u.
- w_i = weighting of the recommendation dimension.
- r_i = normalized compatibility score between 0 and 1.

Safety is not a weighted score. It is a hard eligibility constraint. An activity that violates a user's known restriction must be excluded or require appropriate professional review, regardless of its calculated score.

3.4 Example classification

Example user profile

Illustrative assessment

Primary goal
Weight management

Experience
Beginner

Activity baseline
Low active

Training environment
Home / outdoor

Candidate activity suitability



Illustrative scores, assuming no conflicting safety restrictions.

The system recommends walking and bodyweight training as starting activities. The user can accept, reject or replace activities with other eligible alternatives.

4. Recommendation engine specification

This is the intelligence layer that connects the assessment to the rest of the platform.

4.1 Engine responsibilities

The recommendation engine must:

1. Read the latest completed assessment.
2. Identify the user's primary and secondary goals.
3. Evaluate fitness experience and current activity level.
4. Apply age and health-related eligibility constraints.
5. Evaluate activity preferences and available resources.
6. Select suitable training templates.
7. Build an initial weekly activity schedule.

8. Generate nutrition tracking recommendations.
9. Explain why each activity was recommended.
10. Allow users to customize their plans.
11. Track accepted, rejected and completed recommendations.
12. Adjust future recommendations using performance and feedback.

4.2 Recommendation layers

I recommend dividing the engine into five components.

Component	Responsibility
Goal classification	Determines the user's desired outcomes
Eligibility engine	Excludes unsuitable activities and programs
Activity ranking	Scores and ranks eligible activities
Plan generator	Builds a weekly schedule from selected activities
Adaptation engine	Adjusts plans based on user feedback and progress

This separation is important. Changing the recommendation algorithm should not require rewriting the workout execution engine.

4.3 Recommendation categories

Every completed assessment should generate recommendations in these categories:

A. Primary activities

The main physical activities supporting the primary goal.

B. Supporting activities

Activities that improve complementary capabilities.

C. Recovery activities

Appropriate rest, mobility and recovery recommendations.

D. Nutrition strategy

A general nutrition direction aligned with the user's goal.

E. Habit recommendations

Small, practical behaviours that help the user maintain consistency.

4.4 Weekly plan generation

The plan generator must consider:

- Available training days.
- Session duration.
- Experience level.
- Exercise selection.
- Equipment availability.
- Recovery intervals.
- Activity preferences.
- Health restrictions.
- Training volume and progression.

It should avoid assigning every available day as an intense training day.

A typical beginner's schedule may include:

Day	Session
Monday	Beginner strength
Tuesday	Walking

Wednesday	Recovery
Thursday	Beginner strength
Friday	Rest
Saturday	Walking or cycling
Sunday	Rest

The exact schedule should be generated from the user's assessment and appropriate exercise guidelines, not hardcoded for every beginner.

4.5 Nutrition recommendation logic

Nutrition recommendations should be connected to the user's fitness goals.

Goal	Nutrition approach
Weight management	Sustainable calorie management, balanced meals
Muscle development	Adequate protein and energy intake
Endurance	Appropriate carbohydrate and energy availability
Strength development	Adequate protein, energy and recovery nutrition
General fitness	Balanced and consistent eating habits
Mobility	General nutrition and healthy lifestyle support

For weight-related calculations, use validated equations only when sufficient information is available. Present calorie estimates as estimates, not precise biological measurements.

For minors, avoid automated calorie-deficit and weight-loss prescriptions. Prioritize healthy behaviours and qualified professional guidance.

4.6 Hybrid user control

The recommendation engine must not force users into a rigid fitness plan.

The user should be able to:

- Accept a complete recommended plan.
- Accept selected activities only.
- Replace an activity.
- Change workout days.
- Change workout duration.
- Select an alternative exercise.
- Pause a program.
- Change their primary goal.
- Request a new assessment.

When a user changes their plan, the engine should validate the revised schedule again.

4.7 Continuous adaptation

The engine should learn from three types of feedback:

Feedback type	Example	System response
Explicit	User says a workout is too difficult	Offer an easier eligible alternative
Behavioural	User repeatedly skips evening workouts	Suggest a different time
Performance	User completes a training program consistently	Evaluate readiness for progression

Do not interpret every skipped workout as a loss of motivation. It may reflect time constraints, injury, equipment access or an unsuitable recommendation.

A periodic review should ask:

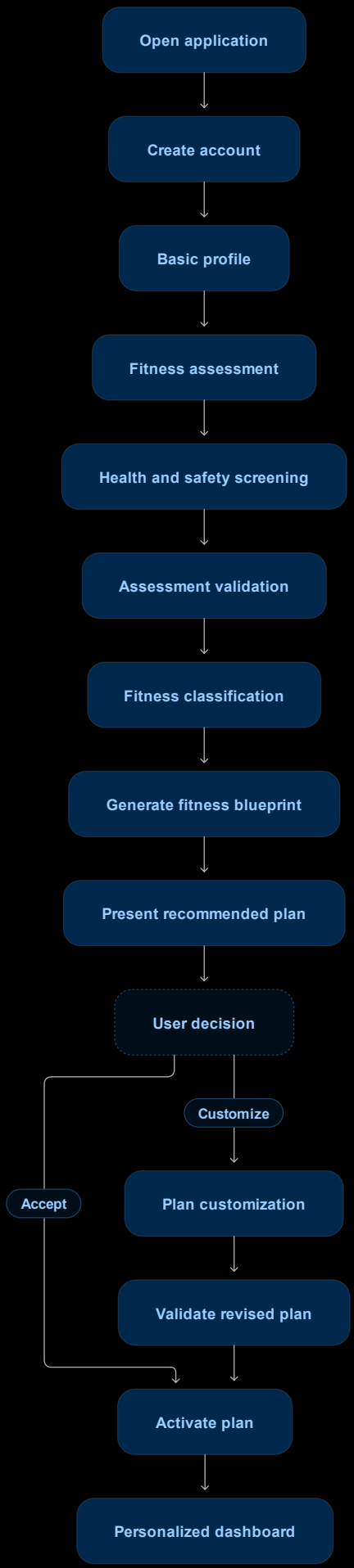
- How manageable is your current plan?
- Are you enjoying your selected activities?
- Have you noticed improvements?
- Is your current schedule still practical?

- Would you like to change your goals?

5. End-to-end user journeys

The product should have a clear user journey from registration to long-term engagement.

Journey 1: New user onboarding



Expected outcome: The user completes onboarding and reaches a personalized dashboard with an actionable plan.

Journey 2: User starts a workout

1. User opens the dashboard.

2. The application displays today's scheduled workout.
3. User views exercise instructions.
4. User starts the session.
5. The application tracks sets, reps, weights and duration.
6. User completes the workout.
7. Performance data is saved.
8. The application updates progress indicators.
9. The recommendation engine receives the completed-session event.

Journey 3: User tracks an outdoor activity

1. User selects running, walking or cycling.
2. The application requests the required location permissions.
3. User starts GPS activity recording.
4. The application records location, duration, distance and pace.
5. User pauses, resumes or ends the activity.
6. The application saves the activity summary and route.
7. The user can share it with friends or a fitness group.
8. The activity contributes to their weekly progress.

GPS tracking must support permission denial, temporary signal loss, interrupted sessions and battery-conscious operation.

Journey 4: User logs nutrition

1. User opens the food diary.
2. User selects a meal.
3. User searches the food database, scans a barcode, uses supported photo-assisted logging or enters food manually.
4. User selects portion size.
5. The application records estimated nutritional values.
6. Daily calorie and macro totals update.
7. The dashboard reflects the nutrition progress.

Journey 5: User changes their goals

1. User opens fitness profile settings.
2. User selects a new primary goal.
3. Application identifies affected training and nutrition recommendations.
4. Recommendation engine generates a revised blueprint.
5. User reviews the changes.
6. User accepts or customizes the new plan.
7. Previous goals and assessment history remain available.

Journey 6: Periodic progress review

1. User receives a scheduled progress review prompt.
2. User reviews completed activities and nutrition consistency.
3. User updates relevant measurements or personal records.
4. Application compares current results with the original baseline.
5. Engine identifies areas for adjustment.
6. Application proposes changes.
7. User approves the updated plan.

6. Product requirements and functional specification

The following requirements define the first product release and subsequent capabilities.

6.1 Functional requirements

ID	Module	Requirement	Priority
FR-001	Authentication	Register, login, logout and account recovery	P0
FR-002	Profile	Create and update user profiles	P0

FR-003	Assessment	Complete multi-stage fitness questionnaire	P0
FR-004	Assessment	Save incomplete assessment progress	P0
FR-005	Assessment	Maintain assessment history	P0
FR-006	Classification	Generate user fitness classification	P0
FR-007	Safety	Apply health and age-based restrictions	P0
FR-008	Recommendations	Generate personalized activity recommendations	P0
FR-009	Recommendations	Generate weekly workout schedules	P0
FR-010	Recommendations	Allow plan customization	P0
FR-011	Workouts	Browse and follow structured programs	P0
FR-012	Workouts	Track exercise sets, reps and weights	P0
FR-013	Activity	Record basic activities manually	P0
FR-014	Nutrition	Search food and log meals	P0
FR-015	Nutrition	Track calories and macronutrients where supported	P0
FR-016	Dashboard	Display goals and daily progress	P0
FR-017	Progress	Maintain workout and activity history	P0
FR-018	Notifications	Deliver workout and progress reminders	P1
FR-019	Activity	GPS tracking and route recording	P1
FR-020	Community	Friends, social feed and interactions	P1
FR-021	Community	Challenges and group activities	P1
FR-022	Progress	Periodic fitness reassessment	P1
FR-023	Recommendations	Adapt plans from user feedback	P1
FR-024	Nutrition	Meal planning and saved recipes	P1
FR-025	Integrations	Wearable and health platform integrations	P2
FR-026	Intelligence	AI-assisted coaching and explanations	P2
FR-027	Monetization	Premium subscriptions	P1

Priority definitions:

- P0: Essential for initial release.
- P1: Important for the first commercial release.
- P2: Advanced capabilities for later releases.

6.2 Non-functional requirements

Category	Requirement
Performance	Standard API requests should target p95 latency below 500 ms, excluding external integrations
Scalability	Support horizontal scaling of stateless services
Security	Enforce authentication, authorization and secure session management
Privacy	Encrypt sensitive data in transit and at rest
Reliability	Prevent loss of workout, assessment and nutrition records
Offline support	Support offline workout logging and queued synchronization
Accessibility	Support accessible controls, readable text and screen readers
Localization	Support local food, currency, units and time zones
Observability	Centralized logs, metrics, traces and error monitoring
Maintainability	Modular services and documented interfaces
Data portability	Allow users to export their personal fitness data

Data deletion	Support account and personal data deletion workflows, subject to applicable legal retention requirements

6.3 Security and privacy requirements

Health-related information requires special treatment.

The platform should implement:

- Explicit consent for sensitive information collection.
- Role-based access control.
- Encryption of sensitive records.
- Restricted access to health-related data.
- Secure token handling.
- Audit logging for sensitive administrative actions.
- User-controlled privacy settings.
- Location privacy and route visibility controls.
- Retention and deletion policies.
- Protection against unauthorized access to user activity data.

For a Nigerian launch, review applicable Nigerian data protection requirements, including the Nigeria Data Protection Act 2023, and determine obligations for international users before expanding.

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7. Technical architecture

Considering the product's different functional domains, I recommend a modular monolith for the initial release, rather than immediately deploying multiple microservices.

This provides clear domain boundaries without introducing unnecessary infrastructure and distributed transaction complexity.

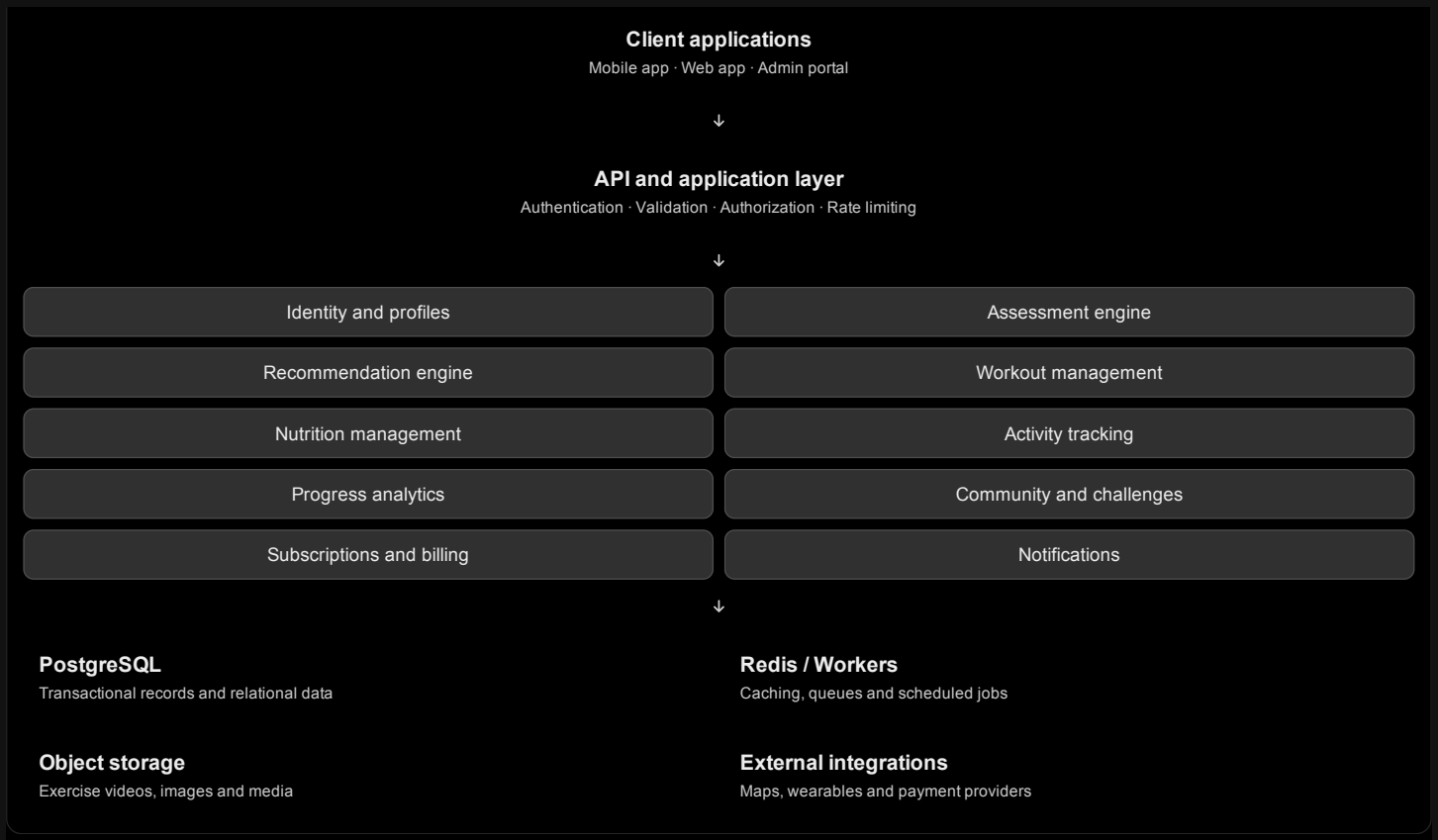
7.1 Proposed technology stack

Layer	Recommended technology
Mobile application	React Native with Expo
Web application	Next.js
Backend	Node.js + TypeScript + NestJS
Database	PostgreSQL
ORM	Prisma
Cache	Redis
Background processing	BullMQ
File and media storage	Cloudflare R2 or S3
Authentication	JWT access and refresh tokens
Payments	Paystack, with other regional providers as required
Maps and GPS	Mapbox or Google Maps
Notifications	Firebase Cloud Messaging and Apple Push Notification service
Analytics	PostHog
Error monitoring	Sentry
Deployment	Docker with a cloud hosting environment
CI/CD	GitHub Actions

This is a proposed stack, not a requirement to lock the product into these providers.

7.2 Backend module boundaries

Application architecture



The recommendation engine should initially be an independent domain module with a versioned interface. It does not need to be a separately deployed microservice.

7.3 Core database entities

Entity	Important fields
`users`	id, email, password_hash, status
`user_profiles`	user_id, date_of_birth, height, weight, preferences
`fitness_assessments`	id, user_id, version, status, completed_at
`assessment_questions`	id, section, question, input_type, configuration
`assessment_options`	id, question_id, value, metadata
`assessment_responses`	assessment_id, question_id, response
`health_restrictions`	user_id, restriction_type, status, notes
`fitness_goals`	user_id, goal_type, priority, target, deadline
`fitness_profiles`	user_id, experience, activity_level, safety_status
`fitness_pathways`	id, name, description, rules
`activity_catalog`	id, category, difficulty, equipment, duration
`recommendation_runs`	id, user_id, algorithm_version, generated_at
`fitness_recommendations`	run_id, activity_id, score, explanation
`workout_programs`	id, goal, level, duration
`workout_sessions`	id, program_id, scheduled_date
`exercises`	id, name, instructions, media_url
`workout_sets`	id, session_id, exercise_id, reps, weight
`nutrition_entries`	id, user_id, food_id, quantity
`food_catalog`	id, name, serving_size, nutrition_data
`activity_sessions`	id, user_id, activity_type, duration, distance

`activity_routes`	activity_id, route_data, visibility
`progress_records`	user_id, metric, value, recorded_at
`user_feedback`	user_id, recommendation_id, feedback_type
`subscriptions`	user_id, plan_id, status, expiry
`notification_preferences`	user_id, channel, preferences

Important implementation rules:

- Store assessment versions and responses immutably after completion. Corrections should create a new version or revision record.
- Store units explicitly and normalize measurements for calculations.
- Store timestamps in UTC and preserve the user's timezone.
- Use database transactions for workout completion and related progress records.
- Use idempotency keys for synchronization and payment-related operations.
- Store GPS data with appropriate privacy controls and retention policies.
- Store nutrition values with their source and confidence level, especially for estimated local foods.
- Keep recommendation explanations and algorithm versions for auditability.

7.4 Proposed API endpoints

All endpoints should be versioned, authenticated where required, validated and documented.

Assessment

```http
POST   /api/v1/assessments
GET    /api/v1/assessments/current
GET    /api/v1/assessments/:id
PUT    /api/v1/assessments/:id/responses
POST   /api/v1/assessments/:id/complete
```

Fitness profile

```http
GET    /api/v1/fitness/profile
PATCH  /api/v1/fitness/profile
GET    /api/v1/fitness/goals
POST   /api/v1/fitness/goals
PATCH  /api/v1/fitness/goals/:id
```

Recommendations

```http
POST   /api/v1/recommendations/generate
GET    /api/v1/recommendations/current
GET    /api/v1/recommendations/:id
POST   /api/v1/recommendations/:id/accept
POST   /api/v1/recommendations/:id/reject
POST   /api/v1/recommendations/:id/feedback
POST   /api/v1/recommendations/:id/customize
```

Workouts

```http
GET    /api/v1/workout-programs
GET    /api/v1/workout-programs/:id
POST   /api/v1/workout-sessions
```

```
POST   /api/v1/workout-sessions/:id/complete
GET    /api/v1/workout-history
```

Nutrition

```HTTP
GET    /api/v1/foods/search
POST   /api/v1/nutrition/entries
GET    /api/v1/nutrition/diary
GET    /api/v1/nutrition/summary
```

Activities and progress

```HTTP
POST   /api/v1/activities
GET    /api/v1/activities
GET    /api/v1/progress/summary
POST   /api/v1/progress/measurements
POST   /api/v1/progress/reassessments
```

The GPS tracking implementation should additionally support activity lifecycle operations, location batches and resumable sessions.

7.5 Recommendation output contract

Every recommendation run should return a consistent structure that the mobile and web applications can render.

Example:

```JSON
{
  "assessmentId": "assessment_123",
  "profile": {
    "primaryGoal": "weight_management",
    "experience": "beginner",
    "activityLevel": "low_active"
  },
  "recommendation": {
    "pathway": "weight_management",
    "algorithmVersion": "1.0.0",
    "activities": [
      {
        "type": "walking",
        "reason": "Matches the user's preferred activities and current experience.",
        "priority": "primary"
      },
      {
        "type": "bodyweight_strength",
        "reason": "Supports general strength development.",
        "priority": "supporting"
      }
    ],
    "weeklyTarget": {
      "trainingDays": 3,
      "sessionDurationMinutes": 30
    }
  },
  "requiresProfessionalGuidance": false,
  "userCanCustomize": true
}
```

This is illustrative. The actual response should also include eligibility metadata, safety constraints, schedule details, recommendation identifiers and applicable disclaimers.

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8. User interface and navigation requirements

The application should feel like one connected fitness product, not three products stitched together.

8.1 Primary navigation

I recommend five main navigation destinations.

Navigation	Purpose
Home	Daily overview and recommended activities
Train	Workout programs and exercise execution
Activity	Running, walking, cycling and activity history
Nutrition	Food diary, meal planning and nutrition
Profile	Goals, progress, assessment and settings

Community can initially be accessible through the Home and Activity sections, then receive a dedicated destination if engagement justifies it.

8.2 Home dashboard

The home dashboard should display:

1. Personalized greeting.
2. Today's fitness objective.
3. Daily activity progress.
4. Calorie and macro summary, where applicable.
5. Today's recommended workout.
6. Weekly training schedule.
7. Current streak or consistency indicator.
8. Recent activity.
9. Progress toward primary goal.
10. Community challenges and relevant updates.

The dashboard should be personalized by goal. A runner should not see exactly the same dashboard emphasis as someone following a strength program.

8.3 Fitness blueprint screen

After assessment, the user should receive a dedicated screen containing:

- Fitness profile.
- Primary goal and supporting goals.
- Current fitness baseline.
- Recommended activities.
- Weekly training schedule.
- Nutrition strategy.
- Explanation of recommendations.
- Expected review date.
- Edit and customize controls.

This screen should become the reference point for the user's entire fitness journey.

9. Subscription and monetization requirements

The platform should use a freemium model, with premium capabilities built around ongoing value.

Feature	Free	Premium	Coaching
Initial fitness assessment	Yes	Yes	Yes
Fitness profile	Yes	Yes	Yes
Basic activity tracking	Yes	Yes	Yes
Basic food diary	Yes	Yes	Yes
Basic workout programs	Limited	Yes	Yes

Advanced nutrition insights	No	Yes	Yes
Personalized meal planning	Limited	Yes	Yes
Advanced training programs	No	Yes	Yes
Advanced activity analytics	No	Yes	Yes
Progress reports	Basic	Advanced	Advanced
AI-assisted guidance	Limited	Yes	Yes
Human coach access	No	No	Yes
Group coaching	No	Optional	Yes

Additional revenue opportunities:

- Corporate wellness subscriptions.
- Gym and fitness studio partnerships.
- Paid fitness challenges.
- Premium training programs.
- Nutritionist and personal trainer marketplace.
- Sponsored fitness events.
- Affiliate partnerships for fitness equipment.

Do not rely on advertising as the primary revenue source. It risks degrading the experience before the product has established user loyalty.

10. Product analytics and success metrics

The platform needs to measure whether the assessment actually produces better fitness engagement.

10.1 North Star Metric

Weekly Active Users Completing Meaningful Fitness Activities (WAU-MFA).

A meaningful activity could be:

- Completing a planned workout.
- Recording an outdoor activity.
- Logging a meaningful nutrition entry.
- Completing an approved fitness goal milestone.

Do not count opening the app as meaningful fitness engagement.

10.2 Key performance indicators

KPI	What it measures
Assessment completion rate	Percentage of users completing onboarding
Recommendation acceptance rate	Percentage accepting recommended plans
Plan customization rate	Percentage modifying recommended plans
First workout completion	Percentage completing their first session
Weekly active users	Ongoing engagement
Workout adherence	Planned versus completed sessions
Nutrition logging frequency	Consistency of nutrition tracking
Goal progression	Improvement against user-defined goals
30-day retention	Users returning after 30 days
90-day retention	Long-term engagement
Free-to-paid conversion	Subscription performance
Subscription churn	Paid user cancellations

Suggested initial product targets, to be validated rather than treated as industry benchmarks:

Metric	Initial target
Assessment completion	70%+
Recommendation acceptance	65%+
First workout completion within 7 days	50%+
30-day user retention	30%+
Weekly planned workout adherence	60%+

Also measure recommendation safety, the percentage of users encountering unsuitable recommendations, data synchronization failures and user complaints about inaccurate nutrition estimates.

11. Development roadmap

I would execute the project in four phases.

1. **Phase 1: Fitness foundation**

   MVP

   Estimated: 8–12 weeks, subject to team size and content readiness.

   - Authentication and profiles.
   - Assessment questionnaire.
   - Classification engine.
   - Health and safety rules.
   - Recommendation engine v1.
   - Weekly plan generator.
   - Fitness dashboard.
   - Basic workout tracking.
   - Basic nutrition diary.
   - Activity logging.

   Exit criteria: A new user can complete an assessment, receive a personalized plan and record their first workout.

2. **Phase 2: Engagement and activity**

   Estimated: 6–8 weeks.

   - GPS activity tracking.
   - Activity history and routes.
   - Progress analytics.
   - Workout reminders.
   - Group challenges.
   - Social feed.
   - Progress reassessment.
   - Plan adaptation.

   Exit criteria: Users can consistently follow plans, measure progress and engage with others.

3. **Phase 3: Premium experience**

   Estimated: 6–8 weeks.

   - Premium subscriptions.
   - Advanced nutrition analytics.
   - Advanced training programs.
   - Meal planning.
   - AI-assisted recommendations.
   - Advanced progress reports.
   - Coach integration.

   Exit criteria: Premium users receive clear additional value and the subscription lifecycle works reliably.

4. **Phase 4: Scale and intelligence**

   Ongoing development.

   - Wearable integrations.
   - Advanced performance analytics.

- Adaptive training intelligence.
- Local food database expansion.
- Corporate wellness.
- Fitness marketplace.
- Multi-country localization.

Exit criteria: The product can support different markets, user populations and commercial partnerships.

These are planning estimates, not commitments. GPS reliability, exercise content production, food database quality and health-safety validation can materially affect delivery time.

12. Key risks and mitigation

Risk	Impact	Mitigation
Too many features at launch	Delayed delivery and poor execution	Strict MVP scope
Inaccurate food data	Loss of user trust	Verified database and transparent estimates
Unsuitable exercise recommendations	Potential user harm	Hard safety constraints and expert review
Low user retention	Weak subscription revenue	Habit loops and adaptive recommendations
GPS inaccuracies	Poor activity experience	Background tracking tests and recovery mechanisms
Expensive exercise video production	Increased operating costs	Licensed content and phased production
AI hallucinations	Unsafe or misleading guidance	Constrained AI and reviewed training templates
Poor subscription conversion	Weak monetization	Test premium value propositions
Privacy concerns	Regulatory and reputational damage	Consent, access controls and data minimization

13. Definition of Done for the assessment and recommendation engine

Before declaring this subsystem production-ready, the engineering team must demonstrate the following:

- ◯ Users can complete the questionnaire on mobile and web.
- ◯ Users can save and resume incomplete assessments.
- ◯ Assessment versions are retained.
- ◯ Primary and secondary goals are correctly recorded.
- ◯ User fitness experience is classified.
- ◯ Activity preferences and constraints are considered.
- ◯ Health-related restrictions are enforced.
- ◯ Age-specific assessment rules work.
- ◯ Recommendations are reproducible for the same inputs and algorithm version.
- ◯ Every recommendation has an explanation.
- ◯ Users can accept or customize recommendations.
- ◯ Modified plans are revalidated.
- ◯ Weekly schedules respect availability and recovery.
- ◯ Recommendations update when goals change.
- ◯ User feedback is captured.
- ◯ Sensitive information is properly protected.
- ◯ Recommendation decisions are auditable.
- ◯ Automated tests cover classification, eligibility and recommendation scenarios.

14. Outstanding product decisions

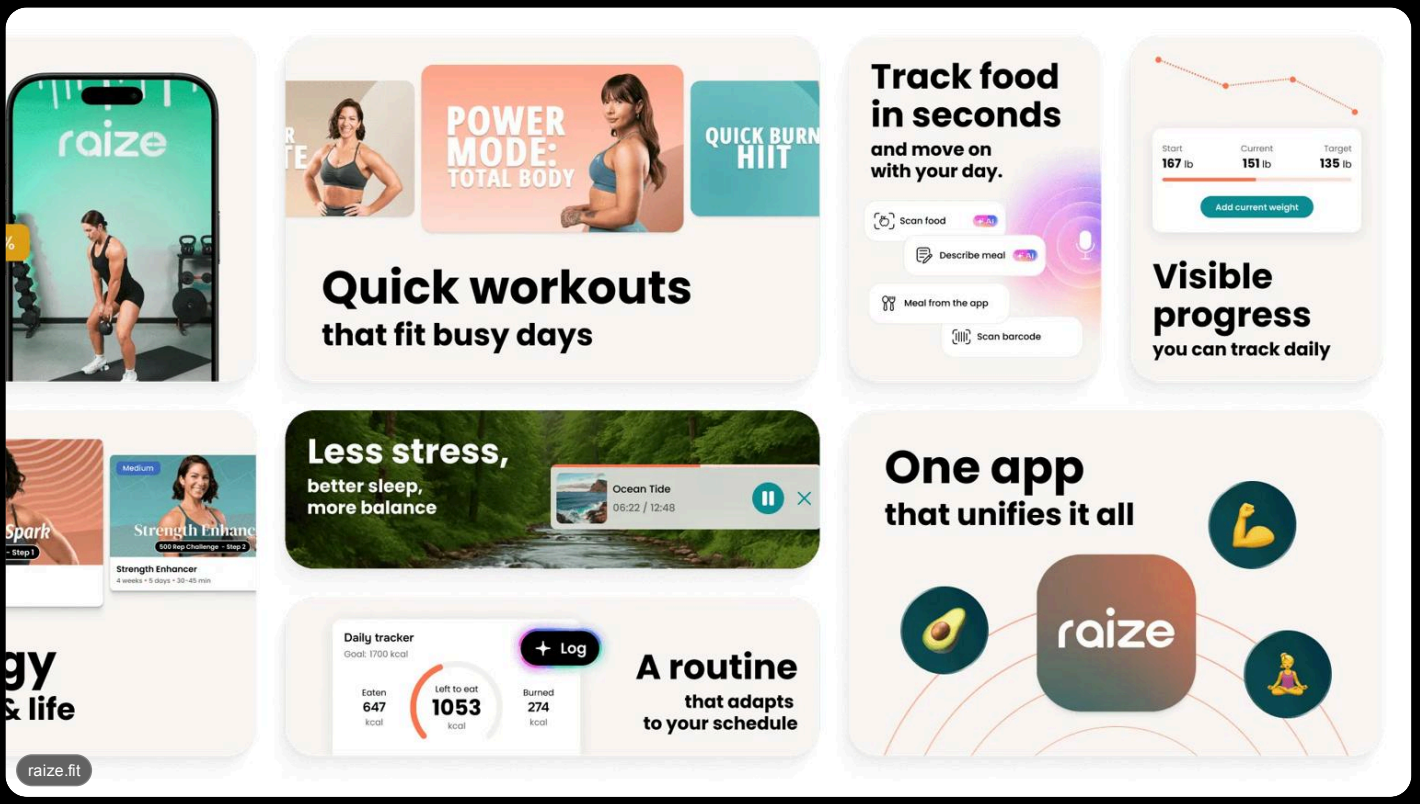
The PRD defines the product direction, but these decisions must be settled before engineering implementation.

Decision	My recommendation
Initial launch market	Nigeria, followed by international expansion

Target age	18+ for initial release; dedicated safeguarded experience for minors later
Initial platform	Mobile first, web dashboard alongside it
Assessment duration	3–5 minutes
Recommendation model	Rules-based scoring with hard safety constraints
AI implementation	Phase 3, with controlled capabilities
Training content	Expert-reviewed exercise library
Food database	International foods plus Nigerian food database
Subscription model	Freemium
Community	Introduce basic challenges in Phase 2
User control	Users can modify their recommended plans
Assessment frequency	Initial, periodic and goal-change assessments

15. Final product direction

One platform. A complete fitness journey.



The product promise

Every user should be able to answer five questions through the platform:

1. Where am I currently?
2. What do I want to achieve?
3. What activities should I engage in?
4. Am I making progress?
5. What should I do next?

My strongest recommendation is to treat the **Fitness Assessment and Recommendation Engine as the product's central intelligence layer**, not an onboarding feature.

MyFitnessPal helps users understand what they eat. Strava helps users understand what they do. LADDER helps users understand how to train.

Your opportunity is to connect all three into a continuous feedback system.

One architectural warning: do not build the AI coach first. Build the exercise catalog, nutrition data, assessment rules, training templates and progress model first. AI without reliable underlying data will produce impressive-sounding but unreliable recommendations.

Recommended next deliverable

The next engineering artifact should be a detailed implementation specification containing:

1. Complete PostgreSQL database schema and Prisma models.
2. Assessment question configuration in JSON.
3. Classification and recommendation rules in TypeScript.
4. Recommendation engine test cases and edge cases.
5. NestJS module structure and API contracts.
6. Mobile and web screen-by-screen wireframe specification.
7. Admin dashboard for managing questions, activities, exercise programs and recommendation rules.

This will turn the PRD into an executable engineering blueprint rather than leaving developers to interpret product intentions independently.